

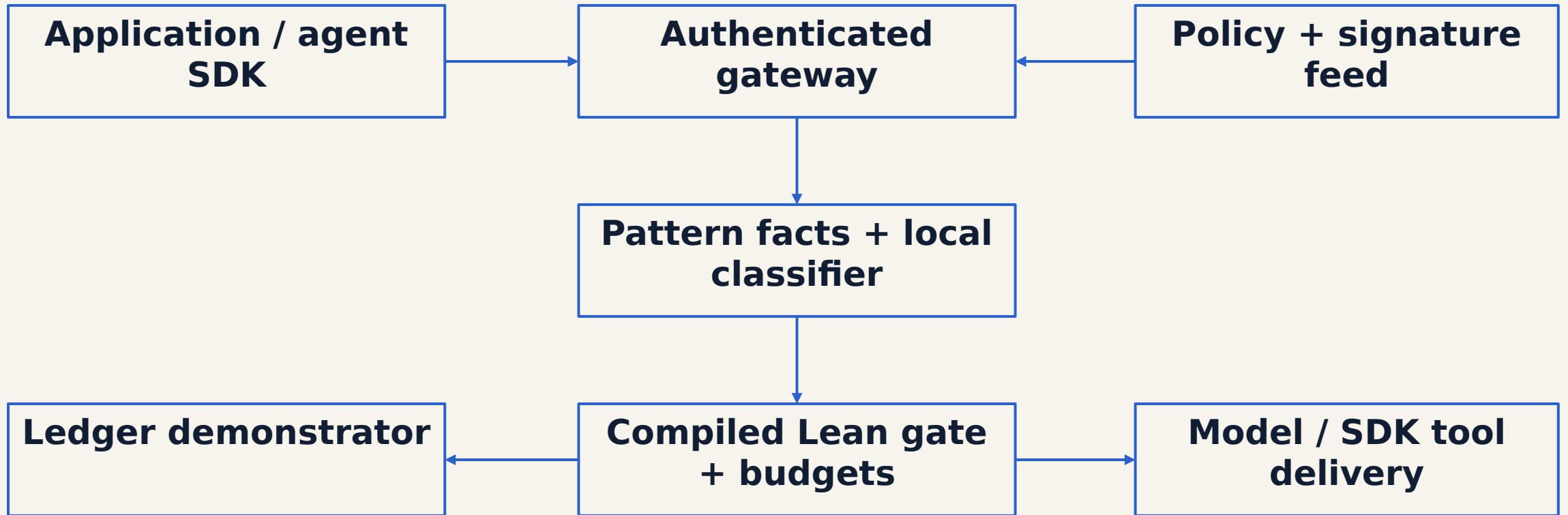
Mathguard

A policy gateway for AI agents

Inspect prompts, tool calls and messages.
Apply policy before releasing content or committing a tool effect.

Team: Mathguard · Goldman Sachs AI Control Layer

One enforcement path



Sanitized decisions and resources feed the dashboard and audit export.

Hybrid decisions preserve hard restrictions

Deterministic facts	Semantic result	Final admission
Signature matched	Safe / allow	Denied
Tool outside allowlist	Safe / allow	Denied
No hard restriction	Above block threshold	Denied + review
Classifier unavailable	Balanced policy	Review; no dispatch

Independent deny, review and redact flags combine by union.

The proofs cover rule composition. Detector completeness remains empirical.

Policy and feed edits take effect

Observed HTTP rehearsal	Active version	Result
Add a synthetic signature	Feed 2	Matching content blocked
Remove that signature	Feed 3	Same content allowed
Tighten PII handling	Policy 8	Email now blocked
Submit invalid policy JSON	Policy 8 retained	Last valid policy kept

Activation preserves ledger balances and previously charged resources.

Resource governance before dispatch

Reserve a configured upper bound

Money, tokens, compute time and call count are charged conservatively.

Stop exhausted budgets and runaway sessions

Actual worker tests reject before dispatch; per-session step and repeat caps apply.

Quarantine after provider timeout

The adapter stops; an operator confirms upstream cleanup before recovery.

Approval and replay protect the ledger

Observed transition	Alice (PLN)	Bob (PLN)
Initial state	1,000.00	200.00
25.00 transfer	975.00	225.00
100.00 approved transfer	875.00	325.00
Exact retries	875.00	325.00

Issuing approval causes no payment. Exact retry causes no new debit or charge.

Attack and artifact boundaries

Indirect and encoded content

Shared controls inspect tool results and session history; encoded sensitive data is refused.

Pinned inert artifacts

Bounded byte/hash/header checks; pickle and code-running formats are refused.

Finite detector coverage

Synthetic feed examples and labeled fixtures demonstrate enforcement paths.

Reproducible evidence

Evidence	Result	Scope
Fresh axiom audit	156 records	Pinned Lean + Mathlib
Gateway/provider/SDK/evaluator suite	83 passed	Actual worker + fixtures
Static audit/parser regressions	14 passed	Provenance and parsing
Public HTTP rehearsal	16 passed	Policy/feed/approval/export

Live-model accuracy and independent unseen evaluation: pending

156 is the audit catalog size, not 156 distinct security requirements.

Reporting for operators and judges

Decision and resource visibility

Dashboard: allow/block/review/redaction, policy/feed version, p50/p95 and remaining bounds.

Sanitized management export

HTTP rehearsal verified that example PII and credentials are absent from exported metadata.

Explicit assurance and current limits

Live versus fixture mode, quarantine, invalid reloads and volatile audit window are visible.

Run, demonstrate, finalize

make test · **make run**

Use the installed local model's exact ID and endpoint.
Record server / weights version and license, then run the live evaluator.

Remaining submission evidence

Live local-model results · independent human-authored prompt set
Browser/operator rehearsal · registered member list · HackTribe entry

github.com/ljaniec/Mathguard/pull/5